

UniEmbedding: Learning Universal Multi-Modal Multi-Domain Item Embeddings via User-View Contrastive Learning

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Abstract

Learning high-quality item embeddings is crucial for recommendation tasks such as matching and ranking. However, existing methods often rely on ID-based item embeddings learned end-to-end with downstream recommendation models, which may suffer from overfitting and limited generalizability. In this paper, we aim to learn universal item embeddings (dubbed UniEmbedding) that capture multi-modal semantics, generalize across multiple domains, and serve different downstream tasks. To achieve this goal, we introduce the UniEmbedding pretraining framework, which includes three modules: a domain-aware multi-modal adapter, a user-view projection module, and contrastive learning objectives across domains. Compared to naive ID embeddings, UniEmbedding provides rich semantic information that generalizes more effectively across domains. Unlike multi-modal embeddings directly extracted from off-the-shelf pretrained models, UniEmbedding achieves better alignment between content semantics and behaviors. We evaluated UniEmbedding on both public and industrial datasets, demonstrating its effectiveness in matching and ranking tasks. Furthermore, UniEmbedding has been deployed in multiple recommendation applications at Huawei, resulting in significant gains in user engagement metrics.

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CCS Concepts

• Information systems → Recommender systems.

Keywords

Item representation, Multi-modal recommendation, Multi-domain recommendation, Contrastive learning, Pretraining

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1 Introduction

In the past decades, ID-based item embedding methods have dominated recommender systems. These methods effectively utilize the co-occurrence relationships between items, leading to the success of numerous classical recommendation models [3, 7, 27]. However, they suffer from overfitting and limited generalizability, particularly when training data is insufficient. Additionally, these methods are highly susceptible to the long-tail effect and struggle to generalize across multi-domain scenarios. As recommendation scenarios become increasingly diverse and complex, ID-based embeddings encounter bottlenecks in scaling across different domains and serving various recommendation tasks. With the rise of powerful pre-trained language models (e.g., BERT [8]) and vision models (e.g., ViT [10]) and their strong generalizability in capturing semantic features, leveraging multi-modal data to enhance the generalizability of recommender systems has become increasingly popular [32].

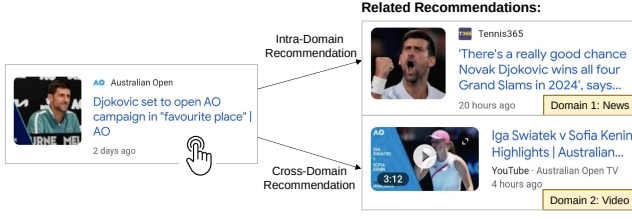


Figure 1: An example of news feed recommendation: A click on a news article will trigger related recommendations, which may contain both news articles (intra-domain) and short videos (cross-domain).

Researchers have explored multi-modal embeddings to address challenges such as the cold-start problem [11] and cross-domain tasks [13, 14, 26]. Towards this goal, some studies focus on creating better multi-modal item embeddings using domain-specific pre-training tasks, while others investigate integrating pretrained item embeddings to enhance downstream models [19].

In this paper, we mainly target the former, aiming to learn universal item embeddings that capture multi-modal semantics, generalize across multiple domains, and serve different downstream tasks. We introduce the UniEmbedding pretraining framework, which includes three designed modules: a domain-aware multi-modal adapter, a user-view projection module, and contrastive learning objectives across domains.

Firstly, different item domains may be dominated by different modalities. For example, as shown in Figure 1, news articles and videos feature different types of content. To unify item representations across domains, we design domain-aware multi-modal adapters to address both their commonalities and unique characteristics. Secondly, existing models such as Item2Vec [3] and ItemSage [2] capture item co-occurrence relationships from a global perspective. However, in practice, item similarities are often characterized by user-specific views. For example, Chinese users frequently purchase spoons and chopsticks together, whereas European users rarely do. To learn universal item embeddings, we decouple an item’s embedding into a global representation and a user-view-specific representation, linked by a user-view projection module. Thirdly, universal item embeddings should generalize across different domains. As the scenario shown in Figure 1, a click on a news article can trigger both intra-domain recommendations for more news articles and cross-domain recommendations for short videos. To achieve this, we need to align item representations across domains. Our work introduces a new multi-domain contrastive loss function that leverages both intra-domain and cross-domain collaborative signals to enhance the pretraining of item embeddings. By leveraging these techniques, UniEmbedding offers significant benefits. Compared to naive ID embeddings, UniEmbedding provides rich semantic information that generalizes more effectively across domains. Unlike multi-modal embeddings extracted from existing pretrained models, UniEmbedding achieves better alignment between content semantics and user behaviors.

We evaluate the performance of our pretrained item embeddings on both public and industrial datasets. UniEmbedding achieves state-of-the-art results in both matching and ranking downstream

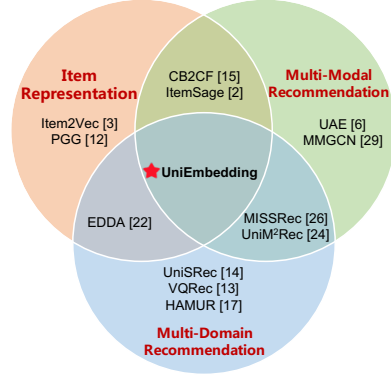


Figure 2: The Venn diagram categorizing different types of recommendation models and pretraining methods.

tasks. We also conducted ablation studies to validate the effectiveness of the user-view projection module and the multi-domain contrastive learning loss. Additionally, UniEmbedding has been deployed in multiple recommendation applications at Huawei, serving the main traffic of users. The contributions of this paper are summarized as follows:

- We propose UniEmbedding, a universal item embedding pre-training framework that demonstrates strong generalizability across multi-modalities and multi-domains, serving matching and ranking downstream tasks.
- We innovatively introduce the user-view projection module and multi-domain contrastive loss function to enhance the performance of pretrained embeddings.
- We conduct extensive experiments and online A/B tests to validate the effectiveness of UniEmbedding.

2 Related Work

Early item embedding pretraining methods are based on item IDs, utilizing the co-occurrence information of items [21] to learn item embeddings, such as Item2Vec [3]. To better leverage the item co-occurrence signals, some ID-based works have built complex item-item graphs [12, 31]. To enable recommendations across multiple domains, some works decouple cross-domain and intra-domain information at the embedding level or model level, thereby enhancing multi-domain knowledge sharing [17, 22].

To align with the emerging trend of multi-modal recommendation, several models for multi-modal item embedding pretraining have been proposed. Due to the semantic gap, directly using multi-modal representations of items in downstream recommendation tasks yields poor results. When pretraining with multi-modal representations, the common approach is to first extract item features using a multi-modal pretraining model [8, 10, 23, 30], then fuse the multiple modalities, and finally these item embeddings are trained under the supervision of users’ collaborative interaction signals. To achieve better recommendation performance, some works use item-item or item-user co-occurrence signals to supervise the adaptation of their multi-modal representations in the recommendation semantic space [2, 6, 15, 29]. To meet the needs of multi-domain

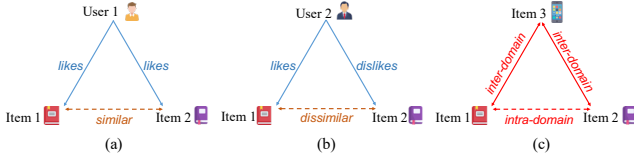


Figure 3: The illustration of the motivation behind UniEmbedding. Figure (a) and (b) illustrate how different samples interfere with the learning of item embeddings. Figure (c) demonstrates cross-domain and intra-domain collaborative signals coexist during multi-domain pretraining.

recommendation, many works leverage the excellent transferability of multi-modal representations, designing sophisticated contrastive loss functions to achieve stronger cross-domain performance [1, 13, 14, 24, 26]. Overall, methods based on multi-modal representations outperform ID-based methods in multi-domain recommendations and addressing long-tail problems.

Figure 2 illustrates a Venn diagram categorizing some of the mentioned works, here, the multi-modal recommendation category is defined as methods that utilize more than one modality, methods that use only one modality, like UniSRec [14], are not included in this category. It is worth noting that, unlike ZESRec [9] and PreRec [18], which focus more on the transferability of the embeddings from the source domain to the target domain, our method emphasizes the consistent improvement of recommendation performance over multiple domains. All above methods struggle to achieve consistent performance across various downstream tasks and domains.

3 UniEmbedding

Figure 3 illustrates the motivation of UniEmbedding. In Figure 3(a), User 1 likes both Item 1 and Item 2, so the embedding vectors of Item 1 and Item 2 should be learned to be more similar. In contrast, according to User 2, the embedding of Item 1 and Item 2 should be dissimilar. This mutual interference between samples makes embedding learning challenging. Therefore, we propose to incorporate user view into the pretraining process to address this issue. In Figure 3(c), three items were liked by a user. Since these items come from two different domains, both cross-domain and intra-domain collaborative signals coexist. Item embedding learning over multiple domains should consider both types of signals.

Our pretraining framework is shown in Figure 4. The training process primarily involves obtaining a user view projection matrix through mixed-domain item sequences interacted with by users. Subsequently, this projection matrix is combined with the domain-adapted embeddings to obtain user-view embeddings. These embeddings are used for contrastive learning pretraining, afterwards. After pretraining phase, the domain-adapted embeddings are applied to downstream tasks because of their fine generalization ability.

3.1 Mixed-Domain Interacted Item Sequence

To ensure that both cross-domain and intra-domain collaborative signals appear in the training data, we utilize the interacted item

sequences from mixed domains for pretraining. Assuming we conduct pretraining on domains A and B, the sequences of user interacted items in these domains are denoted as $S^A = \{I_1^A, I_2^A, \dots\}$, $S^B = \{I_1^B, I_2^B, \dots\}$, where I_i^j represents the i^{th} interacted item in domain j . The input of our framework is the mixed-domain sequence $S^M = \{I_1^M, I_2^M, \dots\}$, where $I_i^M \in S^A \cup S^B$, sorted by user-interacted timestamp, M represents unified mixed domain. For each item $I_i^M = \{e_i, m_i\}$, we use its ID embedding e_i to generate user projection matrix and utilize its multi-modal embedding m_i to obtain the domain-adapted item embedding. The shape of the item embeddings are consistently shaped as $d \times 1$, where the d is the dimension of UniEmbedding.

3.2 Domain-Aware Multi-Modal Adapter

As mentioned in Section 3.1, given item i , its multi-modal embeddings of item sequences, $[m_1; m_2; \dots; m_L]$, will be fed into the domain-aware multi-modal adapters.

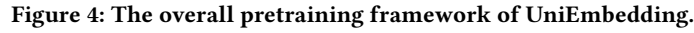
Our adapters adopt the Mixture of Experts (MOE) structure. Each domain has its corresponding experts, and all domains utilize the same shared experts. Domain-specific experts and shared experts are used separately to capture domain-specific information and better leverage shared information of all domains. For each domain, our model has a corresponding gate network, responsible for assigning weights to the outputs of the domain-specific experts and shared experts. Given item i , its unified domain adapted embedding, $E_i^{\mathcal{A}}$ can be calculated through Eqn 1.

$$E_i^{\mathcal{A}} = [\mathcal{E}_1^j(m_i); \mathcal{E}_2^j(m_i); \dots; \mathcal{E}_{n_j}^j(m_i); \mathcal{E}_1^S(m_i); \mathcal{E}_2^S(m_i); \dots; \mathcal{E}_{n_S}^S(m_i)] \cdot \text{Softmax}(\mathcal{G}^j(m_i)), \quad (1)$$

In Eqn 1, j represents the domain of item i , $\mathcal{E}_k^j$ is the k^{th} expert of domain j , n_j is the number of corresponding domain-specific experts. Similarly, $\mathcal{E}_k^S$ represents the k^{th} shared experts, and n_S is the number of shared experts. $\mathcal{G}^j$ corresponds the gate network of domain j . It is worth noting that the L2-normalized $E_i^{\mathcal{A}}$, $E_i^{\mathcal{A}} \mapsto E_i^{\mathcal{A}} / \|E_i^{\mathcal{A}}\|_2$, will be used in downstream tasks introduced in Section 3.3 and Section 3.4. When an item has multiple modality-specific domain-adapted embeddings, the average of these embeddings is used as the embedding for the item.

3.3 User-View Projection

We model user features to generate user projection matrices, obtaining user-view item embeddings. User projection matrices are shaped as $d \times d$, the calculation method of them is shown in Eqn 2. As denoted in Section 3.1, we input the ID embedding sequence $\{e_1, e_2, \dots\}$ into the user behavior encoder to obtain the user behavior representation. Subsequently, the user behavior representation is transformed by the domain-specific multi-head projection layer into the corresponding projection matrix. In Eqn 2, $\mathcal{M}_j^u$, $\mathcal{P}_j$ represents the projection matrix of user u , domain j and the multi-head projection layer, respectively. $\mathcal{T}$ is the user behavior encoder. In our experiment, the encoder consists of a two-layer Transformer [25] encoder blocks. E_i can be calculated as $E_i = E_i^P + E_i^d + e_i$, E_i^P and E_i^d represents position embedding and domain embedding. $[\cdot; \cdot]$ denotes the concatenation operation, L is a hyperparameter denotes max



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Table 1: Matching task performance comparison of different models on Xmarket dataset. UniEmb_{Text} denotes the UniEmbedding method only utilizing item text modality. The overlapped users appeared in the training set, while the unseen users not. The best performance is denoted in bold. The second-best and the third-best performance are denoted underlined fonts.

Test Users	Metric	CLIP	UniSRec	MISSRec	VQRec	Item2Vec	PGG	CB2CF	UAE	UniEmb _{Text}	UniEmb
Overlapped	HR@10	1.458	3.182	3.284	2.637	4.541	<u>4.866</u>	3.225	3.312	<u>6.164</u>	6.661
	HR@50	2.258	3.806	4.301	3.247	7.206	<u>8.134</u>	4.228	4.296	<u>8.520</u>	9.249
	MRR@10	0.824	1.378	1.996	1.623	<u>2.524</u>	2.346	1.984	2.026	<u>3.673</u>	4.248
	MRR@50	0.854	1.440	2.038	1.625	<u>2.640</u>	2.529	2.025	2.065	<u>3.780</u>	4.366
	NDCG@10	0.973	1.802	2.303	1.859	<u>2.998</u>	2.935	2.280	2.332	<u>4.268</u>	4.823
	NDCG@50	1.139	2.139	2.518	1.955	<u>3.573</u>	<u>3.851</u>	2.492	2.539	<u>4.781</u>	5.389
	HR@10	1.238	3.357	3.041	3.347	4.197	<u>5.653</u>	2.979	3.085	<u>6.189</u>	6.570
	HR@50	1.846	4.021	3.994	4.041	6.645	<u>8.332</u>	3.933	4.007	<u>8.588</u>	8.709
Unseen	MRR@10	0.595	1.779	2.069	2.560	2.490	<u>3.319</u>	2.006	2.077	<u>4.203</u>	4.679
	MRR@50	0.606	1.853	2.112	2.579	2.606	<u>3.487</u>	2.049	2.116	<u>4.297</u>	4.782
	NDCG@10	0.743	2.153	2.302	2.745	2.894	<u>3.870</u>	2.239	2.317	<u>4.680</u>	5.131
	NDCG@50	0.853	2.514	2.510	2.878	3.433	<u>4.672</u>	2.447	2.515	<u>5.121</u>	5.606

"electronics", and "cellphones and accessories" domains from the Xmarket dataset in the US market. We filtered users who had interacted with at least two items for the experiment. After filtering, the total number of items is 0.14 million, and the number of users is 2.65 million. Among all users, 65.90% of users only interacted with items from one domain, 30.35% of users interacted with items from two domains, and 3.75% of users interacted with items from all domains.

Industrial Dataset. We conducted experiments using two domains, news article recommendation and short video recommendation. After filtering, there are a total of 1.2 million users, with 40.8% of users interacting with items from only one domain and 59.2% of users interacting with items from both domains. The dataset comprises 0.25 million items and 13.1 million interactions.

For both datasets, we split the training and testing dataset with a ratio of 9:1. In the evaluation phase, we separately tested the performance on overlapped and unseen users. Overlapped users appeared in the training set, and the last interacted item was extracted for prediction. Unseen users didn't appear in the training set.

4.2 Baselines and Metrics

- **UniSRec** [14] is a cross-domain contrastive pretraining method based on the text modality, where the pretraining tasks include seq-seq and seq-item.
- **MISSRec** [26] employs pretraining tasks similar to UniSRec but effectively integrates the image modality. Additionally, MISSRec utilizes clustering methods to model user interests effectively.
- **VQRec** [13] uses a quantization method to map the textual representation of items into a sequence of codes, employing code embeddings to represent items.
- **Item2Vec** [3] is a pretraining method that uses the skip-gram language model to train item embeddings.
- **PGG** [12] is a pretraining method that utilizes item co-occurrence graphs to train item embeddings.
- **CB2CF** [4, 15] is a pretraining method that uses collaborative filtering information to supervise the learning of content-based item embeddings.

- **UAE** [6] applies user-to-item contrastive learning with siamese networks for pretraining.

4.3 Implementation Details and Metrics

For items in Xmarket, we obtain item text by combining the item's domain name, title, categories, and description information with prompt text. Subsequently, we input the text into pretrained CLIP [23] to obtain text embeddings. For the items with images, we input item images into the pretrained CLIP [23] to obtain item image embeddings. For those items without images, we took the item text embedding of CLIP as image embeddings. These embeddings are subsequently fed into the corresponding domain adapters. For the industrial dataset, all items have both text and image information. We input both modality information into the pretraining model FILIP [30] to obtain multi-modal embeddings. Our code was implemented based on RecBole [33] and FuxiCTR [36]. The number of pretraining epochs was set to 300 and the batch size was set to 2048. Our UniEmbedding dimension was 128, and temperature τ was set to 0.07. For the matching tasks, we took HR@K, MRR@K, and NDCG@K, $K \in \{10, 50\}$ as metric. For ranking tasks, we took AUC and logloss as metrics.

5 Results and Discussion

5.1 Evaluation on Downstream Matching Task

Results on Xmarket dataset. The matching task performance on Xmarket is shown in Table 1. For the matching task, we use normalized item embeddings for maximum inner product search to recall items, to align with our production use case for item-item matching. For the baseline methods, as mentioned in Section 2, using the CLIP embeddings without pretraining in the recommendation system performs poorly. The methods mainly based on sequence-item pretraining (i.e., UniSRec, MISSRec, VQRec) exhibit significant performance degradation. The inconsistency between the pretraining task and matching task, as matching tasks use item-item matching, results in suboptimal model performance. Methods based on item ID co-occurrence information (e.g., Item2Vec, PGG) perform the best, likely because the training data is relatively abundant,

Table 2: Matching performance comparison of different models on industry dataset. The best and the second-best results are marked in bold and underlined fonts, respectively.

Test Users	Metric	VQRec	Item2Vec	UAE	UniEmb
Overlapped	HR@10	<u>1.195</u>	1.181	0.526	2.647
	HR@50	3.257	<u>4.481</u>	2.185	8.765
	MRR@10	<u>0.424</u>	0.369	0.162	0.723
	MRR@50	<u>0.512</u>	0.500	0.228	0.983
	NDCG@10	<u>0.603</u>	0.555	0.245	1.164
	NDCG@50	1.045	<u>1.250</u>	0.594	2.475
	HR@10	1.147	<u>1.171</u>	0.527	2.495
	HR@50	3.136	<u>4.408</u>	1.999	8.369
Unseen	MRR@10	<u>0.401</u>	0.348	0.167	0.666
	MRR@50	<u>0.483</u>	0.475	0.227	0.906
	NDCG@10	0.574	0.536	0.249	1.084
	NDCG@50	0.996	<u>1.217</u>	0.560	2.334

allowing the models to capture sufficient item collaboration signals. While CB2CF and UAE utilize both item collaboration signals and multi-modal embeddings, their performance in matching experiments is average. In contrast, our proposed method, UniEmbedding, demonstrates superior performance compared to other baselines for both overlapped and unseen users, even when using only the text modality of items. Moreover, combining the text modality with the image modality further enhances the method’s performance.

Results on industrial dataset. The matching task performance of different baselines on the industrial dataset is shown in Table 2. Due to the significantly high sparsity of the industrial dataset, numerically, all metrics on industrial datasets show a significant decline compared to those on public datasets. However, our approach still maintains a considerable lead over other baselines. Additionally, VQRec performs better on the industrial dataset than on public datasets, indicating that models based on multi-modal embeddings exhibit superior performance when training data is limited.

5.2 Further Analysis

5.2.1 Ablation study. Table 3 shows the ablation experiment results of UniEmbedding on the Xmarket dataset.

w/o User-View. We removed the user behavior encoder, and utilized the mixed-domain-adapted embeddings in Section 3.2 for contrastive learning. We conducted matching experiments using these domain adapted embeddings. The experimental results indicate that when not using user-view embeddings, there is a significant decline in performance. Indicate the effectiveness of our user-view embeddings.

w/o cross-domain. In this experiment setup, we stopped using mixed-domain sequence mentioned in Section 3.1 for pretraining. Instead, each user only utilizes item interaction sequences from a single domain. In this scenario, the model cannot leverage the collaborate signals of items from different domains. It can be observed that the model’s performance declines, indicating the effectiveness of our multi-domain design.

5.2.2 Performance On Long-tail Items. We label the top 50% of items in terms of frequency of occurrence in the training set as head items, and the remaining 50% as tail items. The interactions of tail items account for 10.9% of the total interactions.

Table 3: Ablation study on Xmarket dataset.

Metric	w/o User-View	w/o cross-domain
HR@10	-1.366	-0.168
HR@50	-2.256	-0.171
MRR@10	-0.451	-0.139
MRR@50	-0.498	-0.139
NDCG@10	-0.671	-0.146
NDCG@50	-0.874	-0.147

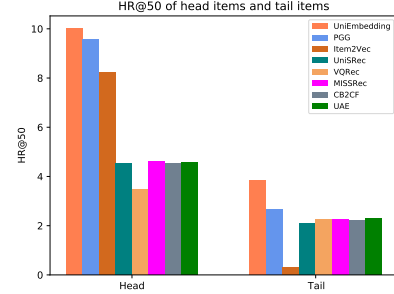
**Figure 6: The HR@50 results on head items and tail items.**

Figure 6 illustrates the performance of different methods on head items and tail items, with the selected metric being HR@50. It can be observed that all methods exhibit a decline in performance on tail items, with PGG and Item2Vec based on only ID information showing the most significant decrease. The other methods utilizing item multi-modal information experience relatively smaller declines. This indicates that utilizing multi-modal information in recommender systems can effectively alleviate the long-tail effect. Our method exhibits superior performance on both head items and tail items, indicating that our approach effectively combines item ID information and multi-modal information.

5.2.3 Performance across multiple domains. We conducted the matching task on items from various domains of public datasets, and the experiment results are shown in Figure 7. The bar chart illustrates the percentage of interactions for each domain out of the total, while the line chart represents the percentage improvement of UniEmbedding compared to the baselines. It can be observed that the baseline only relying on ID signals (PGG and Item2Vec) performs well in domains with sufficient data. Even PGG outperforms UniEmbedding in the largest domain. However, these methods perform relatively poorly in domains with less data. Other baselines utilizing multi-modal signals show relatively consistent performance across multiple domains, indicating that multi-modal embeddings can enhance model performance in multi-domain scenarios. Specifically, VQRec performs well in domains with smaller datasets, highlighting the effectiveness of quantization methods in multi-domain learning. Overall, our UniEmbedding method significantly outperforms all baselines in nearly all domains, demonstrating the effectiveness of our multi-domain design.

5.3 Evaluation on Downstream Ranking Task

We used three powerful ranking models: DCNv2 [28], FinalMLP [20], and DIN [34] to test the benefits of item embeddings obtained from

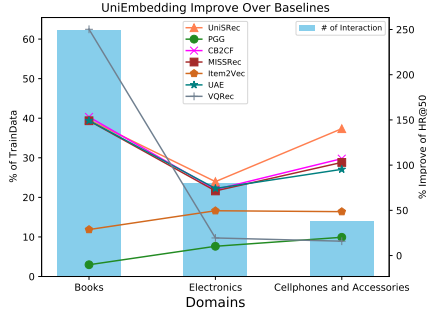


Figure 7: The relative improvement (w.r.t. HR@50) of UniEmbedding compared to all baselines across different domains in the Xmarket dataset.

Table 4: The performance of pretrained embeddings for different ranking models. The best and the second-best performance are shown in bold and underlined fonts, respectively.

Pretrained Embeddings	DCN2		FinalMLP		DIN	
	AUC	Logloss	AUC	Logloss	AUC	Logloss
w/o Emb.	0.8835	0.2842	0.8870	0.2834	0.8896	0.2750
VQ-Rec	<u>0.8985</u>	<u>0.2665</u>	<u>0.8973</u>	0.2673	<u>0.9003</u>	<u>0.2623</u>
UAE	0.8938	0.2740	0.8911	0.2745	0.8976	0.2667
PGG	0.8960	0.2671	0.8967	<u>0.2660</u>	0.8951	0.2697
UniEmb	0.9085	0.2548	0.9010	0.2618	0.9025	0.2617

different pretraining methods on the models. We implemented the models through the open-source code repository FuxiCTR [35, 36]. The input features include the user’s historical interacted item sequence and the corresponding domain sequence, the target item and its domain, and the task is to predict the label indicating whether the user interacts with the target. We conducted experiments on Xmarket dataset, selecting items from the user’s interaction sequence as positive samples, and randomly choose 5 uninteracted items from the same domain as negative samples for each positive sample. The same processing method is used for both the train, valid and test datasets. We concatenated item ID embedding and pre-trained item embedding as model input. The models are trained using early stopping. The ranking experiment results are shown in Table 4. It can be observed that when combined with the UniEmbedding method, all models achieve the best performance. However, when combining UniEmbedding with a stronger ranking backbone (*i.e.*, FinalMLP), the performance is lower than using a less powerful backbone (*i.e.*, DCNv2). It is speculated that the simple concatenation of ID embeddings and pre-trained embeddings may lead to a performance loss. Therefore, when applying UniEmbedding to downstream tasks, it is still necessary to explore better ways of combining pre-trained embeddings to optimize performance. Overall, our method achieved SOTA results in both matching and ranking downstream tasks, indicating the universal applicability of our item embeddings.

5.4 Online A/B Tests

To evaluate the effectiveness of UniEmbedding for industrial recommender systems, we conducted online A/B testing for both matching

and ranking tasks across multiple recommendation scenarios in our production services. The first scenario is the news feed recommendation in the Huawei mobile browser, which serves tens of millions of users daily. The news feed covers two item domains: news articles and short videos, which differ significantly in both content types and user behaviors (with a 1x vs 7.5x difference in CTR). In this setting, we pretrained the UniEmbedding model using historical data from the recent seven days, comprising over 100 million samples from 20 million users. The pretrained 100-dimensional item embeddings were deployed for downstream tasks. We performed online A/B testing using 5% of randomly sampled user traffic for each group. For the matching task, we added a new matching channel based on UniEmbedding for real-time item-to-item recommendations while retaining all other matching channels the same as the base group. Over a period of 10 days, we observed a 1.9% increase in average CTR (click-through rate) on news articles and a boost of 13.9 seconds in average watching time per user on short videos. This improvement underscores the effectiveness of UniEmbedding in enhancing item matching performance.

For the ranking task, we conducted an online A/B test in the recommendation scenario of smartphone lock screen wallpapers. Specifically, we added two embedding features for the downstream ranking model, which include users’ behavior sequences and target items with the corresponding pretrained embedding vectors. We leveraged the soft retargeting network with similarity binning [16] to aggregate the sequence embeddings and further concatenated them into our ranking model (an MMOE-based multi-task model). A 10-day A/B test showed a 2.2% improvement in CTR and a 3.3% improvement in the "like" rate. Currently, UniEmbedding has been employed as an embedding service, serving the main traffic in multiple applications at Huawei.

6 Conclusion

In this work, we propose UniEmbedding, a universal item embedding pretraining framework that demonstrates strong generalizability across multi-modalities and multi-domains, serving both matching and ranking downstream tasks. UniEmbedding innovatively leverages the user-view projection module and multi-domain contrastive loss function to enhance the performance of pretrained embeddings. Experiments on both public and industrial datasets demonstrate the effectiveness of UniEmbedding in both matching and ranking tasks. Additionally, we show the potential of UniEmbedding in addressing multi-domain challenges and improving the representation learning of long-tail items.

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